FNB APP OF THE YEAR HACKATHON 2026 | ACCESS TO HIGHER EDUCATION

ThutoVest

From “I want to study” to “I can graduate.”

A personal plan, early warnings and verified sponsors who pay the university directly. Free for students.

TEAM THUTO

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Scan to try it

thutovest.vercel.app

Thuto means learning in Setswana and Sesotho.
Vest means investing in a student.

IN THIS PRESENTATION

The questions this deck answers

Why	Why does this need to exist?
Who	Who is it for?
What	What is ThutoVest?
How	How does it work, for students and for sponsors?
How	How does it keep students on track after funding?
How	How is it fair, and how are children protected?
How	How does it think, and how is it kept safe?
How	How is it built, and how does it make money?
What	What is real today, and what comes next?
Try	How do I try it in three minutes?

WHY

Matriculants plan their future alone

The challenge: help young South Africans choose a path, fund it, apply and stay supported until graduation.

893,847
NSFAS funding applications for 2026

218,000
still incomplete because documents were missing

Scattered information
A plan pieced together from NSFAS, university sites and WhatsApp groups.

Missed deadlines
Nothing warns them in time, so applications and funding slip away.

The missing middle
Households on R350,000 to R600,000 earn too much for NSFAS, but can’t pay fees.

Bursary scams
Fake offers ask for “admin fees”, and students can’t tell real from fake.

Funded, then dropped
Marks slip quietly. Students hear about it when the funding is already gone.

Sources: NSFAS board chair, reported by TimesLive and EWN on 6 January 2026. Missing middle: households earning R350,000 to R600,000 a year, above the NSFAS threshold. Scams: Daily Maverick, 8 April 2025.

WHO

Built for three people around one student

Students

Grade 11 to university

- Tatenda, Grade 12, 17, Gauteng. A missing-middle family: too much for NSFAS, not enough for fees. Wants Computing and IT, APS 32, a R92,400 gap.
- Nandi, first-year at UJ, funded. Wants to keep her bursary.
- Free, on any phone, with WhatsApp for low data.

Parents and guardians

For learners under 18

- Give consent before any sponsor can see the child.
- Co-sign the bursary agreement.
- Today they do this on the student's phone. Next: their own link, with copies of offers and warnings.
- Nomsa, Tatenda's mother, in the demo.

Sponsors

Companies, trusts and non-profits

- Find verified, ranked students who fit their fund.
- Set their own terms and pay the institution directly.
- See the progress of students they fund.

WHAT

ThutoVest is one place to plan, fund and finish a degree

It turns a student's answers into a personal plan with one next best action, connects them with verified sponsors, and keeps support going once they are studying.

Plan

A personal pathway from choosing a course to first year: course matches, deadlines, documents and Risk Radar warnings.

Fund

The funding gap in rands, NSFAS and other aid, and verified sponsors who pay the institution, never the student.

Stay

Marks analysed every time they are added, advice, tutor requests, and fair early warnings before funding is at risk.

66

qualifications in 8 fields

76

public universities and TVET colleges, with official links

2 sec

for an offer to reach the student's phone

Free for students, always. Sponsors pay a small admin fee on awards paid through the platform.

HOW IT WORKS

Seven stops from matric to graduation

At every stop the app shows one next best action, so the student always knows what to do next.

1

Explore

Courses that fit the APS, field and budget. Change the APS and every match is re-scored.

2

Apply

Official apply links for 26 universities and 50 TVET colleges, with alerts before closing dates.

3

Funding

The gap in rands, and which aid can close it: NSFAS, ISFAP, Funza Lushaka and more.

4

Offers

A verified sponsor finds the anonymised profile and sends an offer, never above the gap.

5

Register

Sign the agreement (a guardian co-signs if under 18) and share documents in one tap.

6

Start

The sponsor approves and pays the institution directly.

7

Stay

Marks are analysed, slips are caught early, and a tutor is one tap away.

Tatenda's example: a missing-middle family, so NSFAS is out and ISFAP is her best public option. APS 32, Computing and IT. The plan shows a R92,400 gap. Ridgeline offers R60,000 for tuition and accommodation, Nomsa co-signs, and R32,400 is left for a second bursary. Other sponsors only see what is left.

HOW IT WORKS FOR STUDENTS

Everything a student needs, in one plan

Study Explorer
66 qualifications in 8 fields, every field with at least six routes, including diplomas and higher certificates.

Risk Radar
Early warnings for deadlines, missing documents and money gaps, each with its reason.

Funding Gap Solver
NSFAS, ISFAP, Funza Lushaka, university aid and loans in one view, with what-if scenarios.

Universities and colleges
All 26 public universities, all 50 public TVET colleges and the DHET private college register, with official links.

Check a place before you pay
NSFAS 2026 accommodation rules and scam signs, before a student pays a deposit.

Document vault
Automatic checks for type, readability and editing. Shared with a sponsor in one tap.

Marks and progress
Every new set of marks is analysed, with advice and a tutor request for funded university students.

Talk to ThutoVest
A WhatsApp bot for students low on data: deadlines, offers and documents.

HOW IT WORKS FOR SPONSORS

Fund the right students, with less admin and no fraud

- Register the organisation. Identity, representative and bank checks before any fund goes live.
- Publish funds with their own terms: what the bursary pays for, a minimum pass mark, work-back years, and whether it must be the only bursary.
- Candidates ranked by fit and by what this sponsor tends to fund, with the reasons shown.
- Offers, document review, funding approval and payment to the institution, never to the student.
- A forecast of the date each fund runs out, and a skills-development spend estimate.

Student progress

For every student they fund, sponsors see each module mark as soon as it is added, flagged when a student slips below the pass mark.

Zanele, first-year at NMU: 56% against a 60% pass mark, Data Structures and Algorithms at 44%, and a tutor request waiting.

One click sends a fair early warning, or arranges the tutor.

HOW A BURSARY FLOWS

From first match to payment, every step checked on the server

1

Match

The student shares an anonymised, verified profile, or the sponsor finds them.

2

Offer

The sponsor offers an amount. The server refuses anything above the gap left, or a third bursary.

3

Sign

The student accepts and signs in the app. Under 18? The guardian co-signs.

4

Documents

Required documents are shared and checked automatically. A child's offer always needs guardian documents.

5

Approve

Payment stays locked until every required document is approved.

6

Pay

The sponsor pays the institution directly. The student's gap drops for everyone.

Safety at every step: a message asking for an "admin fee" is blocked and the sponsor suspended. Real sponsors never ask students for money.

HOW IT KEEPS STUDENTS ON TRACK

Every time marks are added, the app reads and analyses them

Funding is only half the job. ThutoVest catches a slip while there is still time to fix it.

1 Add marks	Type them in, or read them from a report (simulated in this demo). University students add module marks and credits.
2 Analyse	Average, change since last time, APS from NSC levels, slipping subjects and the bursary's pass mark.
3 Advise	On track, keep an eye on it, or at risk, with reasons, the courses now out of reach, and free help.
4 Ask for a tutor	The assistant offers to send a tutor request to the sponsor. One tap to approve. Contact details are removed.
5 Early warning	The sponsor sends a warning the platform words fairly: the requirement, no decision yet, and help.

The warning a student receives

"Your latest average is 56% (Semester 2 tests, 2026), below the 60% this bursary requires. **This is not a decision to stop your funding.** If your results don't satisfy the bursary's requirement at the next review, you could be at risk of losing this funding. Let's make a plan together: reply here, and ask us for a tutor if it would help."

Only the sponsor funding a student sees their marks. A warning note that asks for money is refused.

HOW IT SUSTAINS A STUDENT

From admission to graduation, protecting the funding every year

Students should never lose a bursary out of nowhere. ThutoVest turns a year-end shock into early help.

1

Admission

Registered, and the sponsor has paid the institution. The bursary's conditions, such as a 60% pass mark, are clear from day one.

2

Every test

The student adds marks. The app compares them with last time and with the pass mark, and catches a slip before results are final.

3

Early help

The assistant offers to ask the sponsor for a tutor. The student approves with one tap, and the sponsor arranges it.

4

Early warning

If marks keep slipping, the sponsor sends a fair warning: the requirement, no decision yet, and an offer of help.

5

Every year

Reminders to renew funding and re-register, and a second bursary if the gap grows, within the two-bursary limit.

The cycle repeats every year until graduation.

Without ThutoVest

Marks slip quietly all year. A letter after final results says the bursary has stopped, when it is too late to fix.

With ThutoVest

The slip shows up after the first test. The student gets a tutor, the sponsor gets a plan, and the funding is kept.

HOW IT IS FAIR

Fair funding rules, and children protected by law

Fair funding

- A student holds at most two sponsor bursaries, so one can pay tuition and another the rest.
- Sponsors see the gap left after other bursaries, and no offer or acceptance can go above it.
- An “only bursary” can’t be combined with another.
- Disability-only bursaries match students who say they have a disability. Sponsors see yes or no, never details.
- Money only ever goes to the institution.

Under-18s (POPIA sections 34 and 35)

- Hidden from every sponsor until a parent or guardian consents. The wording and time are recorded.
- Every offer to a child requires the guardian’s consent form, certified ID and proof of guardianship.
- Proof is a birth certificate, or a sworn affidavit if the guardian isn’t a parent.
- The guardian co-signs the bursary agreement.

All enforced on the server. A demonstration build: these legal controls have not been reviewed by a lawyer.

HOW IT THINKS

The app predicts problems before they happen

Explainable rules built on each user's own data. Every alert has a "Why?" that shows the evidence.

IT LEARNS FROM	SO IT CAN
APS, field, budget and province	Re-score every course match the moment the APS changes
Marks after every test or term	Spot a slipping subject, recalculate the APS and warn before a bursary is at risk
The student's pace and activity rhythm	Predict who will fall behind, and nudge before they do
Costs, aid and deadlines	Show the funding gap month by month, with what-if scenarios
A sponsor's past decisions	Rank the students that sponsor is most likely to fund
Fund budget and award pace	Forecast the date each fund runs out
Message text and document contents	Block fee requests, and flag the wrong type, unreadable or edited files

Honest note: these are explainable rules, not machine learning yet. Machine learning comes once there is real outcome data to train on.

HOW IT IS KEPT SAFE

Security built in, and tested against the live app

95
automated checks pass on every change

38 of 38
live attack probes refused

8 of 8
database tables with row-level security

- Every action is checked on the server with the same rules code as the app, so browser and server never disagree.
- Fee and bank-detail requests are blocked and the sender suspended. Sponsors can’t verify themselves.
- Uploads are checked by their real file contents. WhatsApp refuses ID numbers, OTPs, bank details and files.
- Marks go only to the sponsor funding the student. Each demo account resets only its own records.
- Signed webhooks, strict Content Security Policy, HSTS, no secrets in code or history. Demo logins are locked.

HOW IT IS BUILT

One set of rules, on the phone and on the server

On the phone

- Web app: React 19, TypeScript, Vite and Tailwind
- Works on any phone or laptop
- WhatsApp: Talk to ThutoVest for low data

In between

- Vercel serves the app with strict security headers
- Twilio forwards WhatsApp messages as signed webhooks
- Data and logins over HTTPS

On the server (Supabase)

- Auth: real logins with bcrypt-hashed passwords
- Platform function: the app's own rules check every action
- Postgres with row-level security on every table
- Realtime: changes reach open screens in about 2 seconds

Without the backend settings, the same app still runs fully in the browser, so the demo never depends on one service.

WHAT WE BUILT

Built during the hackathon, 23 to 26 October

The student planning core existed before the event. These were built and posted in our Sonke progress updates.

Live backend

Real logins, server-side rules, row-level security and live sync.

WhatsApp bot

Live on real phones through Twilio, opened from the app.

Real institution data

All 26 universities, 50 TVET colleges, the DHET register and NSFAS residence rules.

Fair funding rules

Bursary terms, at most two bursaries, and no offer above the gap left.

Children protected

Guardian consent recorded, and guardian documents on every offer to an under-18.

Marks and progress

Analysis on every upload, advice, tutor requests and fair early warnings.

More study routes

66 qualifications, at least six in every field, including diplomas.

Security hardening

38 of 38 live attacks refused, 95 automated checks, demo logins locked.

HOW IT MAKES MONEY

Free for students, paid for by the funding it unlocks

How it earns

- Free for students, always.
- A small admin fee on awards paid through the platform, charged to the sponsor’s award, never to the student.
- Impact and skills-development reports for sponsors.
- Cohort insights for schools and institutions.

What comes next

- Real identity and bank verification partners
- USSD menus for phones with no data
- NSFAS status and institution confirmations
- Email and SMS alerts, and personal WhatsApp updates
- Forecasts trained on real outcomes
- Real sign-up on the server for new students and sponsors

WHAT IS REAL

What is real today, and what is simulated

Real

- Live app with real logins and live sync between devices
- Server-side rules and row-level security
- WhatsApp bot on real phones (Twilio sandbox)
- Official university and college data, with sources and dates
- Document type and tamper checks, scam blocking
- Bursary, child-safety, marks and progress rules, enforced on the server

Simulated, and labelled in the app

- Identity, guardian and bank verification
- Payments to institutions and institution confirmations
- Sponsors, students and awards: fictional
- Fees, APS guides and some deadlines: demo figures
- Reading marks from a report: the student checks them
- The Navigator chat is scripted; WhatsApp shows the demo student

A demonstration build. The POPIA and children’s controls were designed by the team from the Act and have not been reviewed by a lawyer.

QUESTIONS JUDGES ASK

What, why and how, answered

How is this different from NSFAS or a bursary website?

It joins the whole journey: plan, apply, fund and stay on track, with one next action at every step. Sponsors pay institutions, not students.

Why would a sponsor use it?

Verified, ranked candidates, their own terms enforced, progress of the students they fund, and impact reports. Less admin, no fraud.

How do you stop scams?

Messages are screened. Fee and bank requests are blocked and the sender suspended. Money only goes to institutions.

How do you protect children?

Under-18s stay hidden until a guardian consents. Guardian documents and a co-signature on every offer.

How do you stop students losing funding?

Marks are read after every test. The assistant offers a tutor request, and the sponsor can warn early. No surprises at year end.

How does a parent see what is happening?

Today on the student's phone: consent and co-signing. Next: a guardian link with copies of offers and warnings.

What if a student has no data?

WhatsApp works today. USSD menus are next.

Is the intelligence real AI?

Explainable rules from each user's own data, with the evidence shown. Machine learning once real outcomes exist.

TRY IT

Try it in three minutes at thutovest.vercel.app

Use two screens: one tab signed in as a student and one as the sponsor, side by side on the same computer (each tab keeps its own login), or two phones. Every password is Demo#2026.

Tatenda, Grade 12, 17	**tatenda@thutovest.demo**
One login per judge	**judge1 to judge5@thutovest.demo**
Nandi, funded first-year at UJ	**nandi@thutovest.demo**
Ridgeline Technologies (sponsor)	**funding@ridgeline.demo**

WhatsApp: send **join salmon-castle** to **+1 415 523 8886**, then **hi**.

The three-minute demo

1. Tatenda: approve Share my profile with Ridgeline.
2. Ridgeline: open Candidates and make an offer.
3. Tatenda: review, sign and accept. Nomsa co-signs.
4. Ridgeline: approve documents, then pay the institution.
5. Nandi: Marks and progress, Use a sample report, Save and analyse. On the dashboard, approve the assistant's tutor request.
6. Ridgeline: Student progress, send Zanele an early warning.

TEAM THUTO

Every student deserves a plan, a fair chance at funding, and support until graduation.

Samalani Maluwa

Reshoketswe Makhura

Tebogo Sasa

Thank you. Try it now.

thutovest.vercel.app

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